

QUADRATIC EQUATION

A quadratic equation is an equation in which the highest power of the variable is two. The general form of a quadratic in the variable x is of the form

$$ax^2 + bx + c = 0.$$

Where a , b , and c are constants and $a \neq 0$. In general, a quadratic equation has two solutions, called roots which may or may not be equal.

* The Zero Factor Principle

The principle states that if $x \times y = 0$, then $x = 0$ or $y = 0$ or both $x, y = 0$. For example,

If $(x-1)(x+2) = 0$ then

$$x-1 = 0 \text{ or } x+2 = 0.$$

* Types of Quadratic Equation

These are;

(i) pure quadratic equation

(ii) Affected quadratic equation

[1(i)]

A quadratic equation of the form

$$ax^2 + bx + c = 0$$

is called a pure quadratic equation if $a \neq 0$ and $b = 0$. Example; $x^2 - 25 = 0$

In solving pure quadratic equations, we use Difference of two squares i.e

$$x^2 - y^2 = (x - y)(x + y).$$

Examples

- ① Find the roots of $x^2 - 25 = 0$
- ② Find the two possible values of x that satisfy the equation $9x^2 - 6 = 10$

A quadratic equation of the form

$$ax^2 + bx + c = 0 \text{ is called}$$

affected quadratic equation, if $a \neq 0$ and $b \neq 0$.

Examples are

$$5x^2 - 6x = 0, 2 - 3x - x^2 = 0, x^2 + 3x + 4 = 0.$$

The methods for solving affected quadratic equations are;

- ① Factorization method
- ② Completing the square method
- ③ Quadratic formula method
- ④ Graphical method.

[100]

Exercises

1 Solve the following equation

a $(m-3)(m+4) = 0$

b $(p-2)^2(p+5) = 0$

c $x(x+7)(x+1) = 0$

d $(w+3)(w-7)(w+8) = 0$

e $g^2(g-2)^2 = 0$

f $x(2-x)^2(3+x) = 0$

(4) Test whether the following equations can be factorised

(i) $x^2 + 5x + 6 = 0$

(ii) $6x^2 + 5x - 6 = 0$

(iii) $10x^2 - 13x - 3 = 0$

(iv) $3x^2 - 4x + 5 = 0$

2 (a) $2(x+2)(x+7) = 0$

(b) $2x^2 + 5x = 0$

(c) $7x^2 = 3x$

(d) $6x^2 = 7x + 20$

(e) $2x^2 - 11x + 5 = 0$

(f) $(x+5)(x+3) = 20$

(5) Solve the following exponential equations

(i) $3^{2x} + 2(3^x) - 3 = 0$

(ii) $49^{x+1} - 8(7^{x+1}) + 7 = 0$

(iii) $2^{2x-3} - 2^{x-2} - 1 = 0$

(6) solve

$x - 8\sqrt{x} + 15 = 0$

3 solve $x^2 + \frac{5}{4}x = \frac{3}{2}$

Note

To determine whether $ax^2 + bx + c = 0$ can be factorised, we first evaluate $b^2 - 4ac$.

If $b^2 - 4ac$ is a perfect square, then $ax^2 + bx + c = 0$ can be factorised into a product of two linear factors otherwise, it cannot be factorised.

Nature of Roots

The roots of the quadratic equation

$$ax^2 + bx + c = 0 \text{ where } a \neq 0$$

are given by the quadratic formula as

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-b \pm \sqrt{D}}{2a}$$

$$\text{Where } D = b^2 - 4ac$$

The expression $b^2 - 4ac$ is called the discriminant.

The nature of roots examines the type of roots the equation has. It varies whether the roots are real or imaginary, real or irrational, equal or unequal. The value of the discriminant $D = b^2 - 4ac$ determines the nature of the roots of a quadratic equation.

Thus, If a, b, c are real

1 When $D = b^2 - 4ac > 0$ (i.e. positive) the roots are real and unequal.

Case 1

When $D = b^2 - 4ac$ is a perfect square, the roots are real and rational.

Case 2

When $D = b^2 - 4ac$ is not a perfect square, the

roots are real and irrational. They are conjugate pairs.

2 When $D = b^2 - 4ac = 0$, the roots are real and equal, each being equal to $-\frac{b}{2a}$. Thus the quadratic equation $ax^2 + bx + c = 0$ is a perfect square if $D = 0$

3 When $D = b^2 - 4ac < 0$ (i.e. negative) the roots are imaginary (complex).

Example 1

Examine the nature of the roots of the equations

(a) $2x^2 - 7x + 3 = 0$ (b) $x^2 - 5x - 2 = 0$

(c) $4x^2 - 4x + 1 = 0$ (d) $2x^2 + 2x + 3 = 0$

Solution to be discussed in the class

Exercise

- 1 Find the positive values of m for which the equation $(5m+1)x^2 - 8mx + 3m = 0$ has equal roots
- 2 If the equation $4x^2 + p(x+1) + 5 = 0$ has equal roots. Find the possible values of constant p .
- 3 For what values of p does the equation $x^2 - (4+p)x + 9 = 0$ have real roots

Geometrical Representation of a Quadratic Function

Consider the quadratic function

$$y = ax^2 + bx + c$$

the graph of this function is called a parabola. By method of completing the square

$$y = ax^2 + bx + c$$

$$= a \left[x^2 + \frac{b}{a}x + \frac{c}{a} \right]$$

$$= a \left[x^2 + \frac{b}{a}x + \frac{c}{a} + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} \right]$$

$$= a \left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} + \frac{c}{a} - \frac{b^2}{4a^2} \right]$$

$$= a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} \right]$$

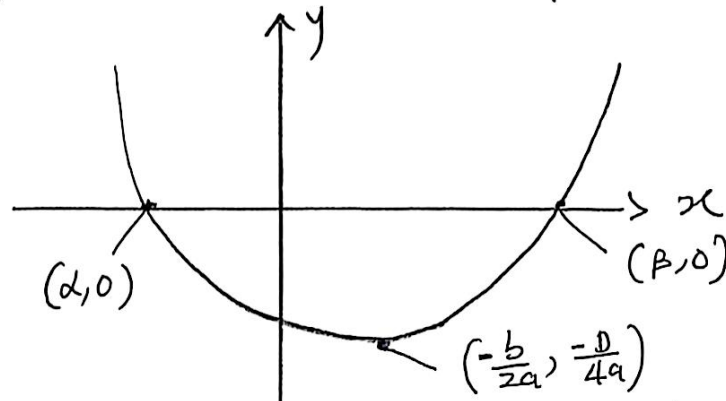
$$= a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{(b^2 - 4ac)}{4a^2} \right]$$

$$= a \left(x + \frac{b}{2a} \right)^2 - \frac{(b^2 - 4ac)}{4a}$$

$$= a \left(x + \frac{b}{2a} \right)^2 - \frac{D}{4a}$$

The maximum or minimum of any quadratic function $y = ax^2 + bx + c$ is given by y-coordinate vertex $-\frac{D}{4a}$ of the corresponding parabola and it occurs at $x = x$ -coordinate of vertex $-\frac{b}{2a}$

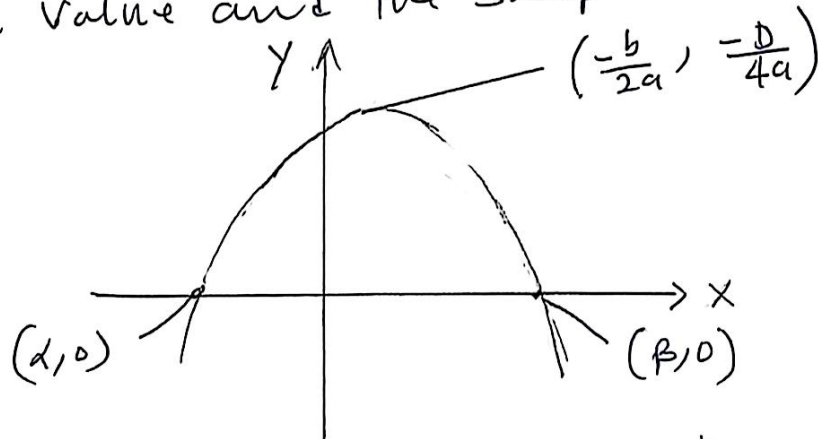
If $a > 0$, the quadratic function $y = ax^2 + bx + c$ has a minimum value and the shape



As $y = a(x + \frac{b}{2a})^2 - \frac{D}{4a}$, $y_{\min} = -\frac{D}{4a}$

when $(x + \frac{b}{2a})^2 = 0$ i.e. $x = -\frac{b}{2a}$

If $a < 0$, the quadratic function $y = ax^2 + bx + c$ has a maximum value and the shape



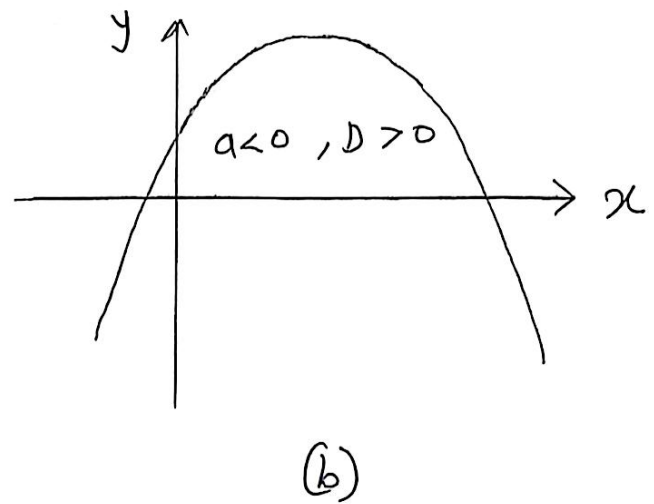
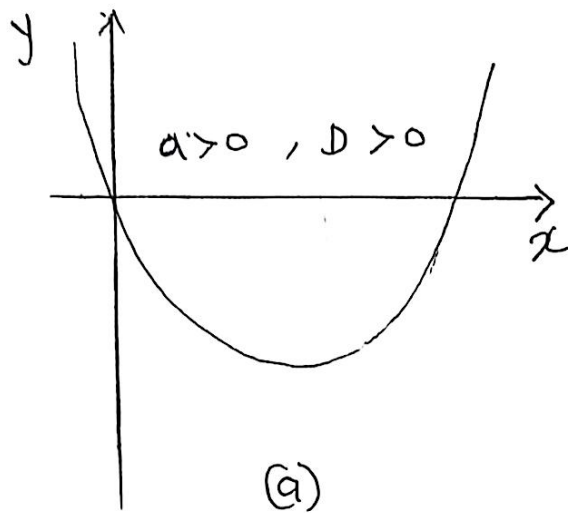
As $y = a(x + \frac{b}{2a})^2 - \frac{D}{4a}$, $y_{\max} = -\frac{D}{4a}$ when

$(x + \frac{b}{2a})^2 = 0$ i.e. $x = -\frac{b}{2a}$

The curve $y = ax^2 + bx + c$ is symmetrical about the line $x = -\frac{b}{2a}$ which is called the axis of symmetry.

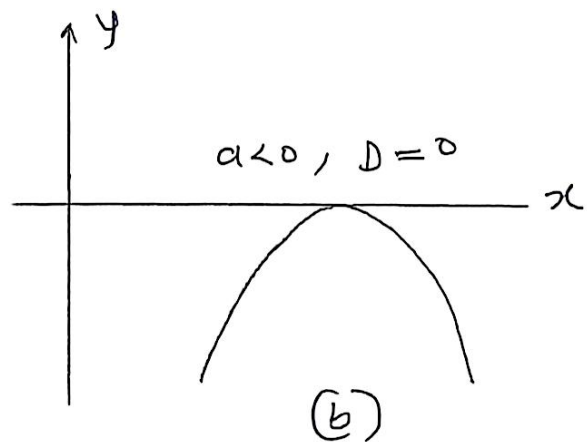
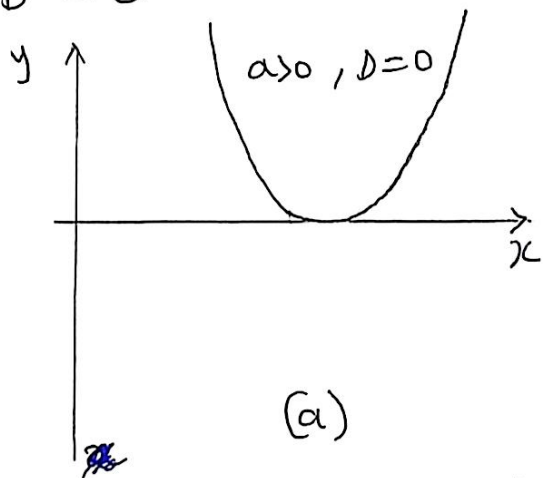
If $y = 0$, then

- 1 The curve cuts the x -axis if $D > 0$

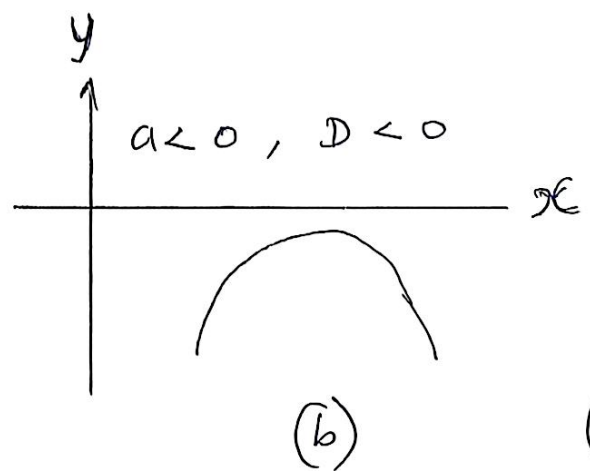
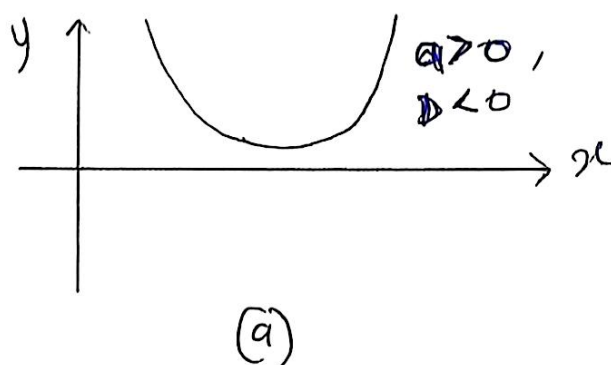


- 2 The curve just touches the x -axis at one point

if $D = 0$



- 3 The curve does not intersect the x -axis if $D < 0$. In this case, the curve will remain either completely above the x -axis or below the x -axis according as $a > 0$ or $a < 0$ respectively.



Illustrative Examples

1. Find the maximum value of the function $-8 + 2x - 4x^2$ and the corresponding value of x .

Solution

$$\begin{aligned}\text{Let } y &= -8 + 2x - 4x^2 \\ &= -4x^2 + 2x - 8 \\ &= -4\left(x^2 - \frac{1}{2}x\right) - 8 \\ &= -4\left(x^2 - \frac{1}{2}x + \frac{1}{16} - \frac{1}{16}\right) - 8 \\ &= -4\left[\left(x - \frac{1}{4}\right)^2 - \frac{1}{16}\right] - 8 \\ &= -4\left(x - \frac{1}{4}\right)^2 + \frac{4}{16} - 8 \\ &= -4\left(x - \frac{1}{4}\right)^2 + \frac{1}{4} - 8 \\ &= -4\left(x - \frac{1}{4}\right)^2 + \frac{1-32}{4} \\ &= -4\left(x - \frac{1}{4}\right)^2 - \frac{31}{4}\end{aligned}$$

y is maximum when $\left(x - \frac{1}{4}\right)^2 = 0$, i.e. $x = \frac{1}{4}$

Maximum value of $y = -\frac{31}{4}$ //

ALITER 1

$$y_{\max} = \frac{-D}{4a} \quad \text{From } -8 + 2x - 4x^2$$

$$a = -4, \quad b = 2, \quad c = -8$$

$$\text{Then } y_{\max} = \frac{-(b^2 - 4ac)}{4a}$$

$$= -\frac{(2^2 - 4(-4)(-8))}{4(-4)} = \frac{-(4 - 128)}{-16} = \frac{-124}{-16}$$

(7)

$$y_{\max} = -\frac{31}{4}$$

Also, the equation of the line of symmetry is $x = -\frac{b}{2a}$ i.e. y is maximum at $x = -\frac{b}{2a}$

$$\text{Which is } x = -\frac{b}{2a} = -\frac{(2)}{2(-4)} = \frac{-2}{-8} = \frac{1}{4} //$$

ALITER 2

y is maximum at $x = -\frac{b}{2a} = \frac{1}{4}$

$$\begin{aligned} \therefore y_{\max} &= -8 + 2\left(\frac{1}{4}\right) - 4\left(\frac{1}{4}\right)^2 \\ &= -8 + \frac{1}{2} - 4\left(\frac{1}{16}\right) \\ &= -8 + \frac{1}{2} - \frac{1}{4} \\ &= \frac{-32 + 2 - 1}{4} \\ &= -\frac{31}{4} // \end{aligned}$$

Further Exercises

- 2 Find the minimum value of $y = 3x^2 + 5x - 2$ and the corresponding value of x for which y is a minimum.
- 3 Given that $x = (2x^2 - x + 8)y$. Show by the method of completing the square that

$$\left(x - \frac{1+y}{4y}\right)^2 = \frac{1+2y-63y^2}{16y^2}$$

- ④ Find the maximum and minimum value of $x^2 - 4x + y^2 + 3 = 0$ if x and y real number satisfying the equation.

Roots of Quadratic Equation

Consider for $a \neq 0$

$$ax^2 + bx + c = 0 \quad \text{————— ①}$$

Let α and β be the roots of ①, so that $(x - \alpha)$ and $(x - \beta)$ are the factors of ①. Hence

$$\begin{aligned} ax^2 + bx + c &= (x - \alpha)(x - \beta) \\ &= x^2 - (\alpha + \beta)x + \alpha\beta \end{aligned}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = x^2 - (\alpha + \beta)x + \alpha\beta$$

Comparing the coefficients, we have

$$(\alpha + \beta) = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

To find the values of the symmetric function of α and β

- i Find the values of $\alpha + \beta$ and $\alpha\beta$ from the given equation
- ii Express the given function ^{in terms} of $\alpha + \beta$ and $\alpha\beta$
- iii Substitute the values of $\alpha + \beta$ and $\alpha\beta$ from (i).

Note: The values of α and β should not be determined separately.

Example 1

If α and β are the roots of the equation $ax^2 + bx + c = 0$, obtain in terms of a , b and c the value of

(i) $\alpha^2 + \beta^2$

(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$

(iii) $\alpha^3 + \beta^3$

(iv) $\alpha - \beta = \frac{b}{a}$

Solution

To be discussed in class.

Example 2

- 1 If one root of equation $px^2 + qx + r = 0$ is three times the other roots, show that $3q^2 = 16pr$.
- 2 The roots of the equation $x^2 + px + q = 0$ are γ and σ . Form the quadratic equation whose roots are $\gamma + \sigma$ and $\frac{1}{\gamma} + \frac{1}{\sigma}$.

POLYNOMIAL

A polynomial in x is an expression involving power of x , normally arranged in ascending or descending powers. The degree of the polynomial is given by the highest power of x occurring in the expression e.g.

- 1 $5x^4 + 7x^3 + 3x - 4$ polynomial of 4th degree Quartic
- 2 $2x^3 + 4x^2 - 2x + 7$ polynomial of 3rd degree Cubic
- 3 $3x^2 + 4x + 2$ Polynomial of 2nd degree Quadratic
- 4 $2x - 3$ polynomial of 1st degree - Linear

Evaluation of Polynomials by Nesting

Consider the following polynomial

$$\begin{aligned} 1 \quad f(x) &= 5x^3 + 2x^2 - 3x + 6 \\ &= (5x^2 + 2x - 3)x + 6 \\ &= [(5x + 2)x - 3]x + 6 \end{aligned}$$

$$\begin{aligned} 2 \quad F(x) &= 2x^4 - 5x^3 - 3x^2 + 4 \\ &= (2x^3 - 5x^2 - 3x + 0)x + 4 \\ &= [(2x^2 - 5x - 3)x + 0]x + 4 \\ &= \{[(2x - 5)x - 3]x + 0\}x + 4 \end{aligned}$$

$$3 \quad F(x) = 4x^3 + 3x^2 + 2x - 4$$

$$\begin{aligned}
 F(x) &= 4x^3 + 3x^2 + 2x - 4 \\
 &= (4x^2 + 3x + 2)x - 4 \\
 &= [(4x + 3)x + 2]x - 4
 \end{aligned}$$

IF $f(x) = 3x^4 + 2x^2 - 4x + 5$, Evaluate $f(2)$

$$\begin{aligned}
 F(x) &= (3x^3 + 0x^2 + 2x - 4)x + 5 \\
 &= [(3x^2 + 0x + 2)x - 4]x + 5 \\
 &= \{[(3x + 0)x + 2]x - 4\}x + 5 \\
 &= \{[(3(2) + 0)2 + 2]2 - 4\}2 + 5 \\
 &= [(14)2 - 4]2 + 5 \\
 &= (24)2 + 5 \\
 &= \underline{\underline{53}}
 \end{aligned}$$

Evaluate (1), (2), (3) for $f(3)$

Remainder Theorem

The remainder theorem states that, if a polynomial $f(x)$ is divided by $(x-a)$, the quotient will be a polynomial $g(x)$ of one degree less than the degree of $f(x)$, together with a remainder R still to be divided by $(x-a)$.

$$\frac{f(x)}{x-a} = g(x) + \frac{R}{x-a}$$

$$\text{So } f(x) = (x-a) \cdot g(x) + R.$$

When $x=a$

$$f(a) = 0 \cdot g(a) + R = f(a) \in \mathbb{R}$$

$$\therefore f(x) = f(a)$$

i.e. If $f(x)$ were to be divided by $(x-a)$, the remainder would be $f(a)$.

Exercise

$$\begin{array}{c} \text{Dividend} \\ f(x) = \text{Divisor} \times \text{Quotient} + \text{Remainder} \end{array}$$

- 1 What is the remainder when $3x^3 - 2x + 3$ is divided by $2x+1$?
- 2 What is the value of k in $x^3 + kx - 3$ if $x-3$ is a divisor?
- 3 Solve $(x^3 + 3x^2 - 13x - 15) \div (x-3)$. Ans (241) (243) (245)

Apply the remainder theorem to determine the remainder

- (4) $(5x^3 + 4x^2 - x + 1) \div (x-2) \Rightarrow 175$
- 5 $(4x^3 - 2x^2 + 5x - 3) \div (x-4) = 241$
- 6 $(x^3 - 2x^2 - 3x + 5) \div (x-5) = 165$
- 7 $(2x^3 + 3x^2 - x + 4) \div (x+2) = 2$

Factor Theorem

IF $f(x)$ is a polynomial and on substituting $x=a$ gives a remainder of zero i.e. $f(a)=0$ then $(x-a)$ is a factor of $f(x)$

For example if

$$f(x) = x^3 + 2x^2 - 14x + 12$$

$$= (x^2 + 2x - 14)x + 12$$

$$= [(x+2)x - 14]x + 12$$

for $x=2$

$$f(2) = [(2+2)2 - 14]2 + 12$$

$$= [8 - 14] \times 2 + 12$$

$$= -12 + 12 = 0$$

So, that the division of $f(x)$ by $(x-2)$ gives zero remainder, i.e. $(x-2)$ is a factor of $f(x)$. The remaining factor can be found by long division of $f(x)$ by $(x-2)$.

That is

$$\begin{array}{r} x^2 + 4x - 6 \\ x-2 \overline{) x^3 + 2x^2 - 14x + 12} \\ \underline{\ominus x^3 \oplus 2x^2} \\ 4x^2 - 14x \\ \underline{\ominus 4x^2 \oplus 8x} \\ -6x + 12 \\ \underline{\ominus -6x \oplus 12} \\ 0 \end{array}$$

$$\text{If } f(x) = x^3 - 5x^2 - 2x + 24$$

$$= (x^2 - 5x - 2)x + 24$$

$$= [(x-5)x-2]x + 24$$

$$f(3) = [(3-5)3-2]3 + 24$$

$$= 3(-8) + 24$$

$$= -24 + 24 = 0$$

By long division

$$\begin{array}{r} x^2 - 2x - 8 \\ x-3 \overline{) x^3 - 5x^2 - 2x + 24} \\ \underline{\ominus x^3 \ominus 3x^2} \\ -2x^2 - 2x \\ \underline{\ominus -2x^2 \oplus 6x} \\ -8x + 24 \\ \underline{\ominus -8x \oplus 24} \\ 0 \end{array}$$

$$\therefore f(x) = (x-3)(x^2 - 2x - 8)$$

$$= (x+2)(x+3)(x-4)$$

④ Factorise completely

(i) $x^3 + 6x^2 + 5x - 12$

(ii) $3x^3 - 4x^2 - 24x - 16$

(iii) $4x^3 - 39x + 35$

~~(iv) $6x^3 + 37x^2 + 67x + 30$~~

(v) $2x^4 - 5x^3 - 15x^2 + 10x + 8$

(vi) $2x^4 + 5x^3 - 20x^2 - 20x + 48$

Roots of Polynomial

We shall illustrate this using an example
Solve the equation $2x^3 - 3x^2 - 5x + 6 = 0$

Solution

$$\begin{aligned} \text{Let } f(x) &= 2x^3 - 3x^2 - 5x + 6 - \\ &= (2x^2 - 3x - 5)x + 6 \\ &= [(2x - 3)x - 5]x + 6 \end{aligned}$$

$$\begin{aligned} \text{If } (x-1); f(1) &= [(2-3)1 - 5]1 + 6 \\ &= -6 + 6 = 0. \end{aligned}$$

$\Rightarrow (x-1)$ is a factor.

using long division

$$\begin{array}{r} 2x^2 - x - 6 \\ x-1 \overline{) 2x^3 - 3x^2 - 5x + 6} \\ \underline{\ominus 2x^3 \ominus 2x^2} \\ -x^2 - 5x \\ \underline{\ominus -x^2 \oplus x} \\ -6x + 6 \\ \underline{\ominus -6x + 6} \\ 0 \end{array}$$

$$\begin{aligned} \therefore f(x) &= (x-1)(2x^2 - x - 6) \\ &= (x-1)(x-2)(2x+3) \end{aligned}$$

The zeros of $f(x)$ are

$$x = 1, 2, -\frac{3}{2}.$$

These are the values of x which make $f(x)=0$
and are therefore the roots of the given equation

Partial Fractions

Introduction to Partial Fractions

By algebraic addition

$$\frac{1}{x-2} + \frac{3}{x+1} = \frac{(x+1)+3(x-2)}{(x-2)(x+1)},$$
$$= \frac{4x-5}{x^2-x-2}$$

The reverse process of moving from

$$\frac{4x-5}{x^2-x-2} \text{ to } \frac{1}{x-2} + \frac{3}{x+1} \text{ is called resolving}$$

into Partial Fractions.

In order to solve an algebraic expression -

into partial fraction:

The numerator must be of lower degree than the denominator. ~~But~~ If it is not, then we first divide out

ii Factorise the denominator into its prime factors

Note

Partial fractions is considered preliminary to integrate some functions and in determining the ~~inverse~~ Laplace transform.

Partial Fractions come in three types, and their forms are summarised below.

Type I

Denominator containing linear factors

$$\frac{f(x)}{(x+a)(x+b)(x+c)} = \frac{A}{x+a} + \frac{B}{x+b} + \frac{C}{x+c}$$

Type II

Denominator containing repeated linear factors

$$\frac{f(x)}{(x+a)^3} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3}$$

Type III

Denominator containing quadratic factor

$$\frac{f(x)}{(ax^2+bx+c)(x+d)} = \frac{Ax+B}{ax^2+bx+c} + \frac{C}{x+d}$$

Illustrative examples

Resolve $\frac{11-3x}{x^2+2x-3}$ into partial fractions

2 Convert $\frac{2x^2-9x-35}{(x+1)(x-2)(x+3)}$ into the sum of three

3 Express $\frac{35x-14}{(7x-2)^2}$ in partial fractions.

4 Express $\frac{42x+44}{(6x+5)^2}$ in partial fractions

Resolve

5 $\frac{3x^2+8x+13}{(x-1)(x^2+2x+5)}$ into partial fractions.

(6) Resolve into partial fraction $\frac{2x^2+2x+10}{(x+1)(x^2+9)}$

Exercises

Resolve the following into partial fractions

1 $\frac{x^3-x^2-3x+5}{(x-1)(x^2-1)}$

(7) $\frac{x^2+4x-7}{(x+1)(x^2+4)}$

(13) $\frac{10x+37}{x^2+3x-28}$

(2) $\frac{7-2x}{(x+1)(x-2)^2}$

(8) $\frac{x^2+2}{(x^2+2x+3)(2x+1)}$

(14) $\frac{3x^2-8x-63}{x^2-3x-10}$

(3) $\frac{6x-10}{x^2+2x-3}$

(9) $\frac{x^2}{(x+1)^3}$

(15) $\frac{2x^2+6x-35}{x^2-x-12}$

(4) $\frac{x^3-x^2-4}{x^2-1}$

(10) $\frac{7x+2}{(2x-3)(x+1)^2}$

(5) $\frac{4x+11}{(x^2+4x-5)}$

(11) $\frac{7x-18x-7}{(x-4)(2x^2-6x+3)}$

(6) $\frac{13-5x^2}{(x^2-1)(x+3)}$

(12) $\frac{x+7}{x^2+7x+10}$